



3652 - Age and Confirmation of White Dwarf Exoplanet Candidate WD 0141-675b

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Imaging	(1) WD0141-675
	2		NIRSpec Fixed Slit Spectroscopy	(1) WD0141-675

ABSTRACT

We request 1.7 hours MIRI and NIRSpec observations to confirm, characterize and determine the age of the white dwarf exoplanet candidate WD 0141-675b. The candidate, identified with Gaia astrometry, has a mass of 9 Jupiter masses and an orbital period of 33 days. There is no other existing observatory capable of independently confirming this exoplanet. The planet orbits only 0.17 AU from the white dwarf, close enough that it could not have survived the red giant phase of its host star at its current location without engulfment. How the planet arrived at its current location is a mystery. Given the planet-host separation and planet mass it is nearly impossible to explain with existing white dwarf exoplanet formation theories. We propose to use JWST MIRI and NIRSpec to detect IR excess from WD 0141-675b's thermal emission spectrum. A detection will measure the planet's temperature and probe its atmospheric composition, while a non-detection would refute the existence of the planet. If confirmed, WD 0141-

675b would be the first white dwarf planet with a known mass and temperature, allowing for a precise determination of the exoplanet's age and therefore time of formation providing a pivotal clue to the planet's origin. Either way, JWST observations will help us understand and confirm this unique planet and the mysterious processes that shape planetary systems after the main sequence.

OBSERVING DESCRIPTION

We request 1.4 hours of MIRI imaging observations in seven filters, from 5-24 microns, of the exoplanet candidate host WD 0141-675. We will use a cycling 4-point dither pattern for all our MIRI observations. For all MIRI filters, we adjusted our exposure times such that we are at <60% saturation. Additionally, we request 0.3 hours of NIRSpec G395M observations (3-point dither) to constrain the planet's temperature/cloud cover. There are no scientific requirements for when the observations must take place. We will acquire on the target star.

Proposal 3652 - Targets - Age and Confirmation of White Dwarf Exoplanet Candidate WD 0141-675b

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WD0141-675	RA: 01 43 0.1013 (25.7504221d)	Proper Motion RA: -0.05691198309917954 sec of time/yr	
			Dec: -67 18 46.14 (-67.31282d)	Proper Motion Dec: -1.0185090000959462 arcsec/yr	
			Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[White dwarfs]				

Proposal 3652 - Observation 1 - Age and Confirmation of White Dwarf Exoplanet Candidate WD 0141-675b

Observation	Proposal 3652, Observation 1										Wed Aug 02 00:00:29 GMT 2023
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	WD0141-675	RA: 01 43 0.1013 (25.7504221d) Dec: -67 18 46.14 (-67.31282d) Equinox: J2000			Proper Motion RA: -0.05691198309917954 sec of time/yr Proper Motion Dec: -1.0185090000959462 arcsec/yr Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Star Description=[White dwarfs]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	8	2	1	Dither 1	4	8	188.703	134806
	2	F770W	FASTR1	8	2	1	Dither 1	4	8	188.703	134806
	3	F1000W	FASTR1	8	2	1	Dither 1	4	8	188.703	134806
	4	F1280W	FASTR1	8	2	1	Dither 1	4	8	188.703	134806
	5	F1500W	FASTR1	8	2	1	Dither 1	4	8	188.703	134806
	6	F1800W	FASTR1	10	3	1	Dither 1	4	12	355.205	134806
	7	F2100W	FASTR1	6	27	1	Dither 1	4	108	2086.83	134806
	8	F2550W	FASTR1	5	17	1	Dither 1	4	68	1121.116	134806

Proposal 3652 - Observation 2 - Age and Confirmation of White Dwarf Exoplanet Candidate WD 0141-675b

Observation	Proposal 3652, Observation 2										Wed Aug 02 00:00:29 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	WD0141-675	RA: 01 43 0.1013 (25.7504221d) Dec: -67 18 46.14 (-67.31282d) Equinox: J2000			Proper Motion RA: -0.05691198309917954 sec of time/yr Proper Motion Dec: -1.0185090000959462 arcsec/yr Epoch of Position: 2015.5						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[White dwarfs]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	1 WD0141-675	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	134806	
Template	Slit					Subarray						
	S1600A1					FULL						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	S1600A1	NRSRAPID	13	2	1	NONE	3	6	901.889	134806